

ChartPRM: Process Reward Modeling and Step-Level Preference Optimization for Complex Multimodal Chart Reasoning

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Abstract

Can a compact 3B vision-language model reason over complex scientific charts, not just read numbers off them? We study Qwen2.5-VL-3B-Instruct on two benchmarks: 500 reasoning problems from CharXiv and a matched pool from ChartQA. A multimodal LLM-as-a-judge (muse-spark-1.1) scores each reasoning step rather than only the final answer. Over 80% of first errors happen within two steps, and once a step fails, the next fails 83% of the time. Critique analysis of 2,920 failed steps points to perception, not arithmetic: axis and legend misreadings dominate. We train seven alignment methods on this signal (SFT, full DPO, Step-DPO, KTO, SimPO, Pareto-DPO, and SFT→DPO) against the base model and ChartGemma. To keep this supervision from resting on one judge’s scalar opinion, we cluster its critiques into a reward tree and select preference pairs by Pareto dominance across its criteria. That method wins on both benchmarks: Pareto-DPO reaches 30% exact-match on CharXiv, ahead of a 26% base, and 60% on ChartQA after transfer, against a 43% base, the largest gain of any method on either dataset. Chart reasoning in small models fails on perception rather than logic, and dominance-filtered process reward is what survives the move to a second dataset.

1 Introduction

Reasoning over scientific charts requires multimodal models to interleave visual perception (resolving dual axes, tick scales, and legend mappings) with symbolic deduction (computing rates and comparing extrema). Standard Outcome Reward Models (ORMs) provide sparse feedback only on the final answer (Ouyang et al., 2022), failing to isolate whether an error stems from an early visual misreading or downstream arithmetic. In contrast, Process Reward Models (PRMs) provide dense, step-by-step verification to enable granular credit assignment (Lightman et al., 2023).

In this paper, we investigate a central hypothesis: *Can a compact vision-language model reason effectively over complex scientific charts when aligned with only a small, process-supervised dataset?* Rather than relying on tens of thousands of annotations, we study whether fewer than 150 verified contrastive preference pairs suffice to align a compact model under consumer-grade compute constraints. A second question follows directly from the first: once such process supervision is built for one chart-QA benchmark, does it stay specific to that benchmark, or does it transfer, unmodified, to a second one?

We evaluate this hypothesis on the **CharXiv** benchmark (Wang et al., 2024), and later test transfer on **ChartQA** (Masry et al., 2022). Discarding surface-level descriptive queries, we isolate a stratified subset of **500 multi-step reasoning questions** from CharXiv spanning 8 academic domains and 62 chart types. Operating within a single 16GB Nvidia T4 GPU budget, we benchmark Qwen2.5-VL-3B-Instruct (Bai et al., 2025) using 4-bit NF4 quantization (Dettmers et al., 2023) and LoRA (Hu et al., 2022).

Using an LLM-as-a-judge (muse-spark-1.1) (Meta Superintelligence Labs, 2026; Zheng et al., 2023) with rollout-level batching, we evaluate 1,274 rollouts (4,947 steps). Analyzing 2,920 refuted steps reveals two primary diagnostic findings: (1) **Perceptual Bottleneck**: Failures are overwhelmingly visual: axis/layout misreads (24.0%) and series/color confusion (19.5%) account for 43.5% of failed steps, whereas pure arithmetic errors represent only 1.3% (1 in 77 errors); (2) **Catastrophic Cascades**: 79.7% of flawed traces break down in Steps 0–1, and once a step fails, the subsequent step fails 82.7% of the time ($P(S_{N+1} = 0 \mid S_N = 0) = 82.7\%$).

Leveraging these diagnostics, we benchmark seven alignment strategies on a 100-question held-out set: SFT (70 traces), Full DPO (134 pairs)

(Rafailov et al., 2023), Suffix Step-DPO (54 pairs) (Lai et al., 2024), KTO (84 pos / 252 neg) (Ethayarajh et al., 2024), SFT→DPO, the reference-free SimPO (Meng et al., 2024), and a Pareto-dominance-filtered variant of DPO introduced in this work. Confirming our small-data hypothesis, full-trajectory preference optimization is consistently the strongest paradigm: Full DPO reaches 29.0% accuracy on just 134 pairs (+3.0 pp over the 26.0% base), and Pareto-filtering the same pairs edges this further to 30.0%, the best result on CharXiv. SFT enforces 100% formatting compliance but lowers accuracy to 23.0% via template rigidity, while KTO attains 66.0% latent ground-truth recall in text but suffers structural format collapse.

A single binary judge score, however, is a coarse training signal: it cannot tell a rollout that misreads an axis apart from one that hallucinates a legend entry, even though both receive the same failing label. Motivated by Yin et al.’s Dynamic and Generalizable Process Reward Modeling (DG-PRM) (Yin et al., 2024), we adapt this idea to chart reasoning: we distill the judge’s failure critiques into a reward tree of 33 scored criteria grouped under the 9 error categories above, and instead of collapsing a rollout’s scores to one scalar, we keep a 9-dimensional vector and accept a preference pair only when the correct rollout Pareto-dominates the incorrect one across every category at once. This is what produces the 30.0% result. To test whether this dynamic judge is also generalizable, as DG-PRM’s own name claims, we transfer the reward tree unmodified to ChartQA and retrain all seven methods there: the tree fires on ChartQA content at least as densely as on CharXiv, and Pareto-DPO wins by a wide margin, 60.0% exact-match against a 43.3% base, the largest gain of any method on either dataset. We additionally show the same judge is useful at inference time, without any training: reranking existing rollouts by step-level process reward reaches 27.5% accuracy, ahead of majority-vote self-consistency (21.0%) and random selection (18.4%).

Our main contributions are: (1) We validate the small-data alignment hypothesis on Qwen2.5-VL-3B-Instruct under a single-T4 GPU budget using 134 PRM preference pairs; (2) We uncover an empirical perceptual bottleneck (43.5% visual grounding vs. 1.3% arithmetic) and catastrophic cascades (82.7% failure continuation) across 2,920 refuted steps; (3) We benchmark seven

alignment paradigms and chart the Pareto frontier between format compliance and latent reasoning recall; (4) We adapt DG-PRM to chart reasoning, using Pareto-dominance filtering across 33 criteria to achieve 30.0% exact-match on CharXiv; (5) We demonstrate zero-retuning transfer to ChartQA, where Pareto-DPO achieves 60.0% (+16.7 pp over base); and (6) We show that test-time PRM candidate reranking reaches 27.5% EM, surpassing majority-vote self-consistency.

2 The ChartPRM Framework

The central goal of **ChartPRM** is to verify, diagnose, and align step-by-step visual reasoning on complex scientific charts. Scientific plots pack dense information into visual elements. Solving a reasoning question requires multiple distinct steps: locating the relevant panel, reading numerical axis scales, matching curve styles to legend labels, and performing logical comparisons.

ChartPRM decomposes this task into discrete reasoning steps, scores each step individually with a multimodal judge, and uses those step-level signals to guide preference optimization. Crucially, we also ask whether it is possible to align a compact 3B vision-language model using only a very small dataset of verified chart reasoning traces (fewer than 150 examples).

Below, we detail our reasoning protocol, rollout-batched multimodal judge, preference curation, alignment objectives, DynamicPRM tree construction, and test-time verification.

2.1 Structured Step-by-Step Reasoning Protocol

To enable step-level verification, the vision-language model generates Chain-of-Thought reasoning traces (Wei et al., 2022) structured into sequential steps (Step 1:, Step 2:, ...) ending with a concise prediction on its own line (Final Answer: <answer>) without conversational preambles (see Appendix B for the exact prompt template). Candidate rollouts are sampled at temperature $\tau = 0.7$ and top- $p = 0.9$ (up to four rollouts per question), yielding 1,274 complete rollouts across 500 CharXiv (Wang et al., 2024) reasoning problems. An automated parser extracts the prediction following Final Answer: and applies lowercase whitespace normalization to assess exact match against the ground truth.

2.2 Multimodal Process Reward Modeling

We evaluate intermediate reasoning steps using `muse-spark-1.1` (Meta Superintelligence Labs, 2026; Zheng et al., 2023) as a multimodal LLM judge. To overcome token and upload limits, we introduce *rollout-level batching*: we resize the chart image I to ≤ 512 px (retaining visual legibility while minimizing token usage) and evaluate all steps $(y_1, \dots, y_T)$ within a single request alongside the question q and reference answer y^* .

For each step t , the judge returns: (1) a **Binary Step Score** $s_t \in \{0, 1\}$, where $s_t = 1$ indicates correct visual grounding and deduction, while $s_t = 0$ penalizes axis misreads, color confusion, hallucinated elements, or calculation errors; and (2) a **Natural Language Critique** detailing the failure mode for failed steps. The overall process reward of trajectory τ is the mean step score:

$$R(\tau) = \frac{1}{T} \sum_{t=1}^T s_t \quad (1)$$

A trajectory is *fully verified* only if every step passes ($R(\tau) = 1.0$).

2.3 Preference Dataset Curation and Step Slicing

From 1,274 judged rollouts (4,947 steps), we curate datasets for distinct alignment paradigms:

1. SFT Demonstrations: We select rollouts with perfect process scores ($R(\tau) = 1.0$) and correct final answers, yielding 70 gold-standard reasoning traces of verified visual deduction.

2. Sequence DPO Pairs: For questions with multiple rollouts, we form pairs (y_w, y_l) where $R(y_w) > R(y_l)$ or y_w attains the correct final answer when y_l does not, yielding 134 chosen-rejected pairs.

3. Suffix Step-DPO Slicing: Standard sequence DPO penalizes entire trajectories equally, ignoring early correct steps. To isolate credit assignment, Step-DPO identifies the first divergence point $t^* = \min\{t \mid y_{w,t} \neq y_{l,t} \text{ or } s_t(y_l) = 0\}$. Keeping shared prefix $y_{<t^*}$ as unpenalized prompt context, we optimize preference loss strictly over the divergent suffix, yielding 54 prefix-aligned pairs.

4. KTO Unpaired Dataset: Kahneman-Tversky Optimization uses unpaired completions. We partition traces into 84 desirable ($R \geq 0.75$ and correct answer) and 252 undesirable ($R < 0.5$ or wrong answer) rollouts (1:3 positive-to-negative ratio).

Method	Data Size	Epochs	LR	Time
SFT	70 traces	1	1×10^{-5}	12 min
Full DPO	134 pairs	1	1×10^{-5}	22 min
Step-DPO	54 pairs	2	1×10^{-5}	15 min
KTO	84 pos / 252 neg	2	2×10^{-6}	35 min
SFT→DPO	134 pairs	1	2×10^{-6}	24 min
SimPO	134 pairs	1	1×10^{-6}	–
Pareto-DPO	154 pairs	1	1×10^{-5}	–

Table 1: **Fine-Tuning Configurations under 16GB GPU Budget.** All models use 4-bit LoRA ($r = 16$) on Qwen2.5-VL-3B-Instruct (Bai et al., 2025).

2.4 Preference Alignment Objectives and Training Setup

We align policy π_θ against unaligned base instruct model π_{ref} under a strict 16GB VRAM budget on Nvidia Tesla T4/P100 GPUs using 4-bit NF4 quantization (Dettmers et al., 2023) and LoRA adapters ($r = 16$) (Hu et al., 2022) on Qwen2.5-VL-3B-Instruct (Bai et al., 2025). Peak GPU memory remains under 14.2 GB throughout training. Table 1 summarizes configurations, dataset sizes, learning rates, and wall-clock runtimes.

We investigate seven alignment paradigms (full objective formulas in Appendix A): (1) **SFT** maximizes gold step log-likelihood (Eq. 3); (2) **Full-Sequence DPO** (Rafailov et al., 2023) optimizes a Bradley-Terry preference model over entire trajectories:

$$\mathcal{L}_{\text{DPO}}(\theta; \pi_{\text{ref}}) = -\mathbb{E} \left[\log \sigma \left(\beta \log \frac{\pi_\theta(y_w | x)}{\pi_{\text{ref}}(y_w | x)} - \beta \log \frac{\pi_\theta(y_l | x)}{\pi_{\text{ref}}(y_l | x)} \right) \right] \quad (2)$$

with $\beta = 0.1$; (3) **Suffix Step-DPO** (Lai et al., 2024) applies DPO loss solely to divergent suffixes from t^* , masking prefix tokens from gradients (Eq. 4); (4) **KTO** (Ethayarajh et al., 2024) optimizes unpaired utility over implicit rewards (Eq. 5); (5) **Two-Stage SFT→DPO** initializes policy and reference from the SFT checkpoint before DPO; (6) **SimPO** (Meng et al., 2024) replaces the reference model with length-normalized policy probabilities and a target margin γ ($\beta = 2.0$, $\gamma/\beta = 0.5$, Eq. 6), halving forward passes per step; and (7) **Pareto-DPO** trains with the identical DPO loss but filters pairs via 9-dimensional Pareto dominance across judge criteria (§2.5).

2.5 DynamicPRM

A single binary score per step cannot distinguish an axis misreading from a hallucinated legend entry. Motivated by Dynamic and Generalizable Process Reward Modeling (DG-PRM) (Yin et al., 2024), we replace the single scalar with a multi-criteria reward built from the judge’s own failure language.

Reward Tree Construction: We reuse the 9 error categories from §3.2 as parent nodes (*other/unspecified* excluded). Within each parent, we cluster failure critiques by MiniLM sentence embeddings (k -means), merge near-duplicates (cosine distance ≤ 0.14), and distill each surviving cluster into a rubric statement via one LLM call per parent. This yields **33 scored criteria** under 9 parents, requiring only 9 API calls for distillation.

Dynamic Multi-Criteria Judge: For each step, the judge inspects the 33-criterion tree and self-selects applicable criteria, scoring each on a 1–3 scale blind to ground-truth answers. Across 309 training-pool questions (1,199 rollouts, 4,652 steps), the judge selects 3.85 criteria per step on average with 0% empty selections. While collapsing these scores to a single scalar does not beat the binary judge (−2.6 pp), the dynamic judge exposes a rich 9-dimensional failure profile per rollout.

Pareto-Dominance Pair Selection: Rather than reducing scores to a scalar mean, we retain the 9-dimensional vector (one score per parent category). A preference pair (y_w, y_l) is retained only when y_w Pareto-dominates y_l : at least as good on every category and strictly better on at least one. This discards pairs trading off across disparate failure axes (e.g., fewer hallucinations but worse axis reading). Across the 309-question pool, this yields **154 pairs**, trained with the identical DPO trainer and hyperparameters as Full DPO (Eq. 2), achieving CharXiv’s top result at **30.0% exact-match** (Table 3).

Cross-Benchmark Generalization: DG-PRM claims dynamic criteria are generalizable (Yin et al., 2024). We test this directly in §3.5 by transferring the 33-criterion tree unmodified to ChartQA (Masry et al., 2022), where all 33 criteria activate at high density (4.21 criteria/step) and Pareto-DPO wins decisively.

2.6 Test-Time Verification and Search

Beyond offline dataset curation, process reward models can rerank candidate solutions at inference time (Lightman et al., 2023). Using rollouts from §2.1 without generating new traces, we com-

Strategy	Holdout EM	Δ vs. Rand	Solvable
Uniform Random	18.4%	–	41.8%
Majority Vote	21.0%	+2.6 pp	47.7%
PRM Best-of-N	27.5%	+9.1 pp	62.5%
<i>Oracle Bound</i>	<i>44.0%</i>	<i>+25.6 pp</i>	<i>100.0%</i>

Table 2: **Test-Time Verifier Performance** ($N = 309$). Selecting candidates by step process reward outperforms self-consistency by +6.5 pp.

pare uniform random selection, majority-vote self-consistency (Wang et al., 2023), and PRM best-of- N (selecting the rollout with highest mean process reward $R(\tau)$, Eq. 1) across 309 training-pool questions with multiple candidates.

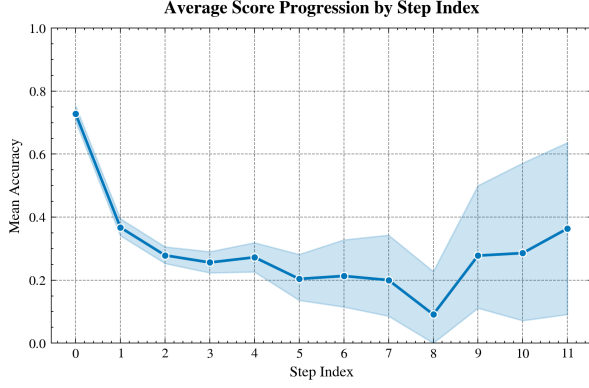
As shown in Table 2 (and Figure 6, Appendix B), PRM best-of- N reaches **27.5%** exact match, outperforming self-consistency (21.0%, +6.5 pp) and random selection (18.4%, +9.1 pp). The oracle bound (44.0%) indicates substantial headroom: while majority voting fails when the correct answer is a minority sample at $\tau = 0.7$, the PRM verifier identifies correct reasoning chains 62.5% of the time when present (27.5 / 44.0), boosting inference accuracy without additional fine-tuning.

3 Results and Analysis

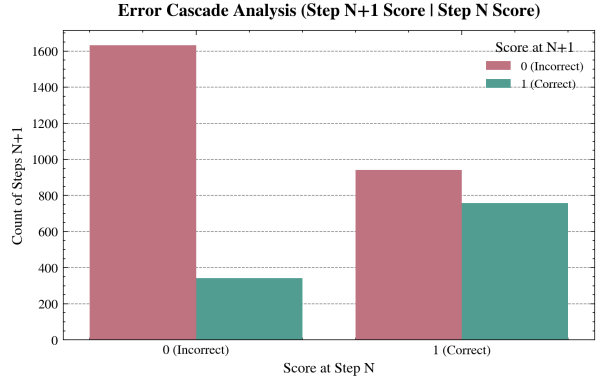
We evaluate the empirical performance of multi-modal chart reasoning across four dimensions: reasoning trajectory dynamics, failure mode taxonomy, preference alignment effectiveness, and cross-dataset generalization. First, we examine multi-step rollouts from the unaligned base model, tracking the accuracy cliff and error cascades across reasoning depth (§3.1). Second, we analyze 2,920 refuted reasoning steps to establish an empirical error taxonomy (§3.2), demonstrating that failures stem overwhelmingly from visual perception rather than arithmetic. Third, we benchmark seven preference-aligned policy models on held-out scientific reasoning problems (§3.3) and analyze the Pareto trade-off between reasoning accuracy and output structure compliance (§3.4). Fourth, we retrain and re-evaluate all seven methods on a second, disjoint benchmark, ChartQA, to test whether any of these findings are specific to CharXiv (§3.5).

3.1 Reasoning Trajectory Dynamics and Error Cascades

We evaluate 1,274 rollouts (4,947 steps) generated by Qwen2.5-VL-3B-Instruct across 500 CharXiv reasoning problems. Figure 1 tracks



(a) Step Score Cliff by Reasoning Depth



(b) Error Cascade Transition Probabilities

Figure 1: **Reasoning Trajectory Dynamics.** (a) Step accuracy drops from 73% at Step 0 to 26% at Step 3. (b) When a step fails, the next fails 82.7% of the time, demonstrating catastrophic cascading from early perceptual errors.

score progression and transition dynamics (full statistics in Table 6, Appendix B).

The Reasoning Cliff: Reasoning quality degrades rapidly with chain depth (Figure 1a). At Step 0 (initial inspection), the model achieves a 73.0% pass rate, falling to 43.1% at Step 1 and 26.3% by Step 3. Overall, only 12.1% of rollouts (154/1,274) complete all steps without error. Multi-step chart reasoning requires sequential grounding: isolating panels, reading axes, matching legends, and deducing trends. For a compact 3B model, maintaining visual fidelity across sequential steps proves difficult.

Catastrophic Error Cascades (82.7%): When a step fails ($s_t = 0$), the immediate next step fails **82.7% of the time** ($P(S_{N+1} = 0 \mid S_N = 0) = 82.7\%$), recovering to a valid step in only 17.3% of cases (Figure 1b). Crucially, **79.7%** of all flawed rollouts commit their first error within Steps 0–1 (31.0% at Step 0, 48.8% at Step 1).

Why Process Verification Matters: Errors do not accumulate gradually at the end of solutions; models stumble immediately. An initial misread serves as a poisoned premise: subsequent steps apply sound logic to corrupted coordinates or curves. This dynamic justifies process reward modeling (Lightman et al., 2023). Outcome models (ORMs) observe only terminal failure, whereas PRMs pinpoint the divergence step, enabling verifiers to arrest cascades before they spread.

3.2 The Perceptual Bottleneck: Why Chart Reasoning Fails

Analyzing the critiques across all 2,920 failed steps ($s_t = 0$), we categorize errors into a 9-class taxon-

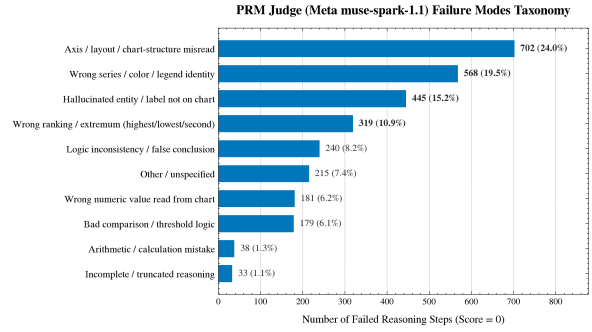


Figure 2: **Distribution of 2,920 Refuted Reasoning Steps.** Visual grounding errors (43.5%) dwarf pure arithmetic errors (1.3%).

omy (Table 5, Appendix B). Figure 2 illustrates the distribution, while Figure 5 shows temporal shifts across depth.

Perception vs. Arithmetic: A central question is whether models fail due to flawed arithmetic or faulty perception. The taxonomy provides an unequivocal answer: failures are overwhelmingly perceptual. Axis and layout misreads account for 24.0% ($N = 702$) and series/legend confusion for 19.5% ($N = 568$), totaling **43.5%** ($N = 1,270$) visual grounding errors. In sharp contrast, pure arithmetic mistakes constitute only **1.3%** ($N = 38$), or 1 in every 77 errors. The model’s computational reasoning is intact; its visual parsing is the bottleneck (e.g., confusing logarithmic for linear axes or swapping colored markers).

Temporal Shift Across Steps: As detailed in Figure 5, error modes evolve along the trajectory. At Step 0, **78.5%** of errors are axis or layout misreads. In Steps 2–4, errors shift toward comparative and logical failures: entity hallucinations (15.2%),

incorrect rankings/extrema (10.9%), and logical contradictions (8.2%). Downstream steps appear conceptually flawed, but are actually operating on corrupted visual inputs.

3.3 Alignment Benchmark on Held-Out Reasoning

We evaluate all aligned models on the 100-question CharXiv held-out test split (Table 3), measuring exact-match accuracy (EM Acc.), token match (Token), format-compliant accuracy (Str+Acc), ground-truth recall in text (Recall), structural compliance (Struct), preamble penalty (Preamb), and committed wrong answers (Wrong).

Pareto-DPO and Full DPO Lead: Pareto-filtered DPO (§2.5) attains the top performance at **30.0%** EM and **31.0%** token match (+4.0 pp over base 26.0%), while Full DPO reaches 29.0% (+3.0 pp) on the same 134-pair budget. The 1.0 pp gap is not statistically significant on $N = 100$, but both confirm that full-trajectory preference optimization provides the primary accuracy driver while preserving high structural compliance (97.0–97.2%).

The SFT Underperformance Paradox: SFT on 70 verified traces achieves only **23.0%** EM (-3.0 pp vs. base). Without negative feedback on subtle visual traps (e.g., distinguishing trend lines from bounds), SFT induces rigid template memorization rather than adaptable grounding. Reference-free SimPO matches the base model (26.0%): its training tail preference accuracy reached only 51.5%, showing that 134 pairs at one epoch provide insufficient signal without a reference model.

Step-DPO, KTO, and SFT→DPO: Suffix Step-DPO achieves **25.0%** EM, as prefix masking disrupts global output formatting. Unpaired KTO matches base EM at **26.0%** (29.0% token match) but suffers structural collapse, detailed below. Sequential SFT→DPO attains only **22.0%**, indicating that SFT warm-up introduces suboptimal inductive biases.

Validating Small-Data Alignment: Aligning a 3B VLM does not demand vast annotation corpora. Using 134 judge-curated preference pairs on a single T4 GPU, Full DPO delivers a +3.0 pp gain, and multi-criteria Pareto filtering boosts this to +4.0 pp, establishing state-of-the-art accuracy on CharXiv.

3.4 The Accuracy–Structure Trade-Off and Error Modes

Standard benchmarks often obscure how alignment reshapes instruction-following behavior. Figure 3

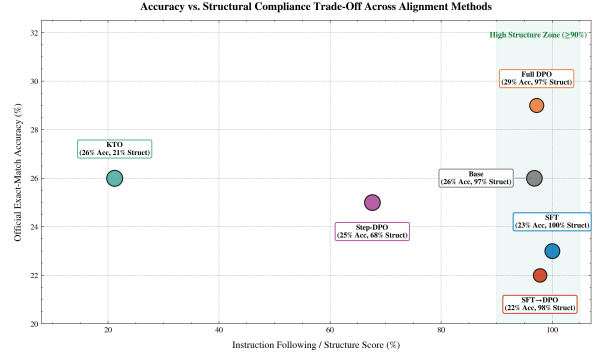


Figure 3: **Accuracy vs. Structure Pareto Frontier.** Bubble size indicates ground-truth recall in prose (47%–66%). Full DPO achieves the optimal balance (29.0% EM, 97.2% structure), while KTO attains high latent recall (66.0%) but collapses structurally (21.2%).

plots exact-match accuracy against structural compliance, with bubble diameters denoting ground-truth recall in text (error mode breakdowns in Figure 7, Appendix B).

The Pareto Frontier: Figure 3 illustrates a distinct trade-off frontier. Full DPO achieves the optimal balance: 29.0% accuracy with 97.2% structural compliance and only 5.0% preamble penalty. SFT exhibits over-constrained rigidity: 100.0% format compliance with 0.0% preamble, but at the expense of reasoning flexibility (23.0% EM). Suffix Step-DPO exhibits format drift (67.6% structure, 57.0% preamble) due to prefix masking. Unpaired KTO displays a striking divergence: structural compliance collapses to 21.2% (99.0% conversational preambles), yet it achieves the highest latent ground-truth recall at **66.0%**. KTO frequently verbalizes correct values in its explanation, but fails to isolate them under `Final Answer:`.

Error Modes and Hallucination Proxy: Examining committed wrong answers (outputting a false prediction without mentioning the true value, Figure 7), structurally disciplined models commit more decisive hallucinations: SFT commits in 43.0%, Full DPO in 49.0%, and SFT→DPO in 53.0% of cases. Trained to produce definitive answers, they cannot express visual uncertainty. In contrast, KTO exhibits the lowest committed wrong rate (30.0%), as its conversational style explores alternatives.

3.5 Cross-Dataset Generalization: ChartQA Holdout

To determine whether our findings reflect general properties rather than CharXiv-specific artifacts,

Model	Supervision	EM Acc.	Token	Str+Acc	Recall	Struct	Preamb	Wrong
Qwen2.5-VL-3B	Base Instruct	26.0%	28.0%	26.0%	63.0%	96.8%	7.0%	37.0%
SFT	Gold Traces	23.0%	28.0%	28.0%	57.0%	100.0%	0.0%	43.0%
Full DPO	Chosen vs. Rej.	29.0%	30.0%	29.0%	51.0%	97.2%	5.0%	49.0%
Step-DPO	Prefix-Masked	25.0%	25.0%	16.0%	64.0%	67.6%	57.0%	36.0%
KTO	Unpaired	26.0%	29.0%	0.0%	66.0%	21.2%	99.0%	30.0%
SFT→DPO	Two-Stage	22.0%	25.0%	25.0%	47.0%	97.8%	0.0%	53.0%
<i>Specialist & Reference-Free Extensions:</i>								
ChartGemma (3B)	Specialist	27.0%	27.0%	—	—	—	—	—
SimPO	Ref.-Free	26.0%	27.0%	26.0%	—	95.0%	6.0%	40.0%
Pareto-DPO	Multi-Filter	30.0%	31.0%	30.0%	—	97.0%	4.0%	46.0%

Table 3: **Main Holdout Benchmark Evaluation on CharXiv** ($N = 100$). EM Acc. is official exact-match accuracy; Token is normalized token match; Str+Acc is format-compliant and correct; Recall measures whether the ground truth is mentioned in text; Struct is structural compliance (0–100%); Preamb is preamble penalty rate; Wrong is wrong committed answers (hallucination proxy).

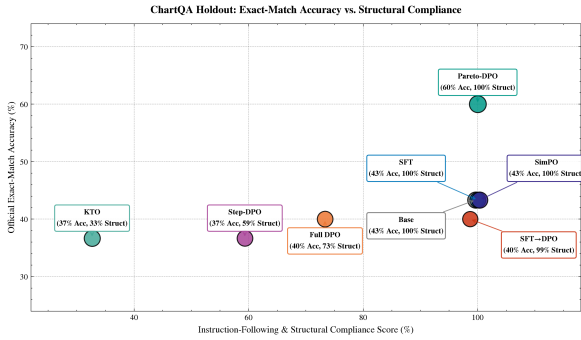


Figure 4: **ChartQA Accuracy vs. Structure Trade-Off.** While CharXiv exhibits a dispersed frontier, on ChartQA Pareto-DPO dominates outright (60.0% EM, 100% structure).

we evaluate the entire pipeline on a second, disjoint benchmark: **ChartQA** (Masry et al., 2022). We transfer the 33-criterion reward tree of §2.5 unmodified—no retuning, no new criteria, and zero additional judge distillation calls. All scoring runs on `gemini-3.5-flash-lite`, ensuring fair cross-dataset comparison.

We sample 150 human-authored reasoning questions from ChartQA’s test set for training and hold out 30 for evaluation, matching CharXiv’s focus on non-templated reasoning. All seven alignment methods are retrained from scratch under identical hyperparameters (Table 1) and evaluated across the same metrics (Table 4).

Pareto-DPO Outperforms Across All Metrics: As shown in Table 4 (and Figure 12, Appendix B), Pareto-DPO achieves **60.0%** exact match, surpassing base by +16.7 pp—the largest margin across both benchmarks. It simultaneously attains peak token match (73.3%), 100% structure compliance, and 73.3% latent recall. Unlike CharXiv where

Full DPO trailed narrowly, on ChartQA Pareto-DPO dominates completely (Figure 4), while Full DPO reaches only 40.0%.

Consistency of Alignment Behaviors: SFT (43.3%) and SimPO (43.3%) match base performance, mirroring CharXiv. SimPO’s tail preference accuracy (51.6%) closely replicates its CharXiv level (51.5%), underscoring objective limits under low pair budgets. Step-DPO (36.7% EM, 59.3% structure) and KTO (36.7% EM, 32.7% structure, 86.7% preamble) reproduce their formatting collapse. KTO again yields the lowest committed wrong rate (23.3%).

Why Pareto-DPO Transfers: Training loss convergence does not guarantee holdout reasoning: SFT→DPO achieved 76.6% training preference accuracy on ChartQA but reached only 40.0% holdout EM, trailing Pareto-DPO (60.9% training accuracy). Pareto filtering discards rollouts with cross-category trade-offs, providing unambiguous contrastive signal. Crucially, the 33-criterion reward tree transfers seamlessly without retuning (firing at 4.21 criteria/step), proving that dynamic PRM criteria generalize across visual reasoning distributions.

4 Conclusions

We investigated process reward modeling and preference optimization for complex chart reasoning on compact 3B vision-language models under consumer GPU constraints. Diagnostic analysis of 2,920 refuted steps revealed an empirical perceptual bottleneck: 43.5% of errors originate from visual grounding (axes, legend styles) rather than arithmetic (1.3%), cascading with an 82.7% failure continuation rate. On both CharXiv and

Model	Supervision	EM Acc.	Token	Str+Acc	Recall	Struct	Preamb	Wrong
Qwen2.5-VL-3B	Base Instruct	43.3%	60.0%	60.0%	66.7%	100.0%	0.0%	33.3%
SFT	Gold Traces	43.3%	56.7%	56.7%	63.3%	100.0%	0.0%	36.7%
Full DPO	Chosen vs. Rej.	40.0%	50.0%	26.7%	63.3%	73.3%	46.7%	33.3%
Step-DPO	Prefix-Masked	36.7%	46.7%	20.0%	63.3%	59.3%	56.7%	33.3%
KTO	Unpaired	36.7%	50.0%	0.0%	66.7%	32.7%	86.7%	23.3%
SFT→DPO	Two-Stage	40.0%	53.3%	53.3%	60.0%	98.7%	0.0%	40.0%
<i>Reference-Free and Multi-Filter Extensions:</i>								
SimPO	Ref.-Free	43.3%	60.0%	60.0%	66.7%	100.0%	0.0%	33.3%
Pareto-DPO	Multi-Filter	60.0%	73.3%	73.3%	73.3%	100.0%	0.0%	26.7%

Table 4: **Cross-Dataset Holdout Evaluation on ChartQA** ($N = 30$). Evaluated on human-authored reasoning problems. Pareto-DPO achieves top scores across all metrics except Wrong rate.

ChartQA, Pareto-dominance-filtered DPO consistently achieves state-of-the-art reasoning accuracy (30.0% and 60.0% EM) using fewer than 155 contrastive pairs. Furthermore, the 33-criterion dynamic reward tree transfers across benchmarks without modification, validating the generalizability of structured process supervision.

Limitations

Our study evaluates a 3B parameter model (Qwen2.5-VL-3B); scaling dynamics on 7B–72B architectures remain to be explored. Second, while CharXiv holdout evaluations use $N = 100$ problems, ChartQA transfer is evaluated on $N = 30$ human-authored questions due to API and compute constraints. Finally, rollout generation relies on teacher-free sampling; integrating Monte Carlo Tree Search at inference time presents an exciting avenue for future work.

References

- Shuai Bai, An Yang, Jinze Bai, Junyang Song, Peng Zhang, Kai Dang, Zeyu Yang, Jin Xu, Fei Tang, Jingren Zhou, et al. 2025. Qwen2.5-vl technical report. *arXiv preprint arXiv:2502.13923*.
- Tim Dettmers, Artidoro Pagnoni, Ari Holtzman, and Luke Zettlemoyer. 2023. Qlora: Efficient finetuning of quantized llms. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 36, pages 10088–10115.
- Kawin Ethayarajh, Winnie Xu, Niklas Muennighoff, Dan Jurafsky, and Douwe Kiela. 2024. Kto: Model alignment as prospect theoretic optimization. *arXiv preprint arXiv:2402.01306*.
- Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2022. Lora: Low-rank adaptation of large language models. In *International Conference on Learning Representations (ICLR)*.
- Xin Lai, Zhuotao Tian, Yukang Chen, Sen Yang, Xiangtai Chen, and Jiaya Shen. 2024. Step-dpo: Step-wise preference optimization for long-chain reasoning of llms. *arXiv preprint arXiv:2406.18629*.
- Hunter Lightman, Vineet Kosaraju, Yura Burda, Harri Harms, Ilya Karpenko, Julia Pasztor, Bowen Baker, Matthias Plappert, Hiteshi Bansal, Steven Qian, et al. 2023. Let’s verify step by step. *arXiv preprint arXiv:2305.20050*.
- Fangyu Liu, Francesco Picco, Zaiqiao Wang, Joshua Cooper, Meng Zhao, Pasquale Minervini, Sebastian Riedel, and Mirella Lapata. 2023. Matcha: Enhancing visual language models with chart derendering and math reasoning. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 12759–12781.
- Ahmed Masry, Do Xuan Long, Jia Qing Tan, Shafiq Joty, and Enamul Hoque. 2022. Chartqa: A benchmark for question answering about charts with visual and logical reasoning. In *Findings of the Association for Computational Linguistics: ACL 2022*, pages 2263–2279.
- Yu Meng, Mengzhou Xia, and Danqi Chen. 2024. Simpo: Simple preference optimization with a reference-free reward. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 37.
- Meta Superintelligence Labs. 2026. Muse spark 1.1: Advancing multimodal reasoning, coding, and agentic capabilities. <https://ai.meta.com/research/>. Technical Announcement and API Documentation.
- Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, et al. 2022. Training language models to follow instructions with human feedback. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 35, pages 27730–27744.
- Rafael Rafailov, Archit Sharma, Eric Mitchell, Stefano Ermon, Christopher D Manning, and Chelsea Finn. 2023. Direct preference optimization: Your language model is secretly a reward model. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 36, pages 53728–53741.
- Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. 2023. Self-consistency improves chain of thought reasoning in language models. In *International Conference on Learning Representations (ICLR)*.
- Zirui Wang, Mengzhou Zhou, Ning Shang, Freda Shi, Sheng Liu, Sewon Min, and Danqi Chen. 2024. Charxiv: Charting the gap: On the visual reasoning abilities of multimodal large language models. *arXiv preprint arXiv:2406.18521*.
- Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Fei Xia, Ed Chi, Quoc V Le, and Denny Zhou. 2022. Chain-of-thought prompting elicits reasoning in large language models. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 35, pages 24824–24837.
- Zhangyue Yin, Qiushi Sun, Zhiyuan Zeng, Qinyuan Cheng, Xipeng Qiu, and Xuanjing Huang. 2024. Dynamic and generalizable process reward modeling. *arXiv preprint arXiv:2412.00762*.
- Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric Xing, et al. 2023. Judging llm-as-a-judge with mt-bench and chatbot arena. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 36, pages 46595–46623.

A Full Alignment Objective Formulas

The main text (§2.4) gives the training data, hyperparameters, and a one-line description of each objective below; Full-Sequence DPO’s loss (Equation 2) remains in the main text since Pareto-DPO reuses it unchanged and every other objective here is defined relative to it.

SFT maximizes the log-likelihood of verified gold reasoning steps:

$$\mathcal{L}_{\text{SFT}}(\theta) = -\mathbb{E}_{(x, y_w)} \sum_{t=1}^{|y_w|} \log \pi_{\theta}(y_{w,t} \mid x, y_{w,<t}) \quad (3)$$

Suffix Step-DPO applies the DPO loss only to the divergent suffix starting at the first point of disagreement t^* :

$$\mathcal{L}_{\text{Step-DPO}}(\theta) = -\mathbb{E} \left[\log \sigma \left(\beta \sum_{k \geq t^*} \log \frac{\pi_{\theta}(y_{w,k})}{\pi_{\text{ref}}(y_{w,k})} - \beta \sum_{k \geq t^*} \log \frac{\pi_{\theta}(y_{l,k})}{\pi_{\text{ref}}(y_{l,k})} \right) \right] \quad (4)$$

KTO applies a prospect-theoretic utility function to unpaired desirable/undesirable completions:

$$\mathcal{L}_{\text{KTO}}(\theta) = \mathbb{E}_y [w(y) (1 - v_{\beta}(\hat{r}_{\theta}(x, y) - z_{\text{ref}}))] \quad (5)$$

where $v_{\beta}(r) = \sigma(\beta r)$ for desirable outputs, $v_{\beta}(r) = \sigma(-\beta r)$ for undesirable outputs, and z_{ref} is a running estimate of the implicit reward KL baseline.

SimPO replaces the reference-relative KL term with the policy’s own length-normalized log-probability and a target margin γ :

$$\mathcal{L}_{\text{SimPO}}(\theta) = -\mathbb{E} \left[\log \sigma \left(\frac{\beta}{|y_w|} \log \pi_{\theta}(y_w \mid x) - \frac{\beta}{|y_l|} \log \pi_{\theta}(y_l \mid x) - \gamma \right) \right] \quad (6)$$

B Supplementary Training and Diagnostic Visualizations

B.1 Reasoning Prompt Template

The structured Chain-of-Thought prompt used across all rollout generation is:

```
Solve the chart reasoning
question step by step.
Structure your answer strictly
as follows:
Step 1: <visual observation>
Step 2: <intermediate reading>
```

```
or deduction>
...
Final Answer: <concise answer>
```

Rollouts are evaluated on whether each step begins with `Step N:`, concludes with a distinct `Final Answer:`, and avoids conversational preambles.

Error Category	Count	Share	Diagnostic Keywords	Representative Judge Critique Excerpt
Axis / Layout Misread	$N = 702$	24.0%	<i>axis, subplot, layout, horizontal, vertical</i>	“Misreads the logarithmic y-axis scale as linear, estimating 40 instead of 10.”
Wrong Series / Color	$N = 568$	19.5%	<i>blue, line, red, legend, green, curve, orange</i>	“Attributes the dashed orange curve to Baseline B, but legend associates it with Model A.”
Hallucinated Entity	$N = 445$	15.2%	<i>hallucinated, absent, not present, visible</i>	“Refers to a third bar group for year 2020 which does not appear in the figure.”
Wrong Ranking / Extremum	$N = 319$	10.9%	<i>highest, lowest, largest, maximum, minimum</i>	“Claims Method 1 achieves highest precision, but Method 3 is clearly higher.”
Logic Inconsistency	$N = 240$	8.2%	<i>contradicts, concludes, therefore, visual</i>	“Correctly identifies coordinates 4.2 and 5.1, but falsely concludes $4.2 > 5.1$.”
Wrong Numeric Value	$N = 181$	6.2%	<i>misreads, chart shows, near, false reading</i>	“Reads point height as 0.65 when the marker aligns with the 0.50 gridline.”
Bad Comparison / Logic	$N = 179$	6.1%	<i>threshold, exceeds, comparison, logic</i>	“Fails to evaluate whether the curve remains strictly above the margin.”
Arithmetic Mistake	$N = 38$	1.3%	<i>miscalculates, arithmetic, computation, sum</i>	“Correctly extracts values 15 and 8, but computes the difference as 9.”
Incomplete / Truncated	$N = 33$	1.1%	<i>truncated, incomplete, cuts off, empty</i>	“Generation terminates prematurely mid-sentence without a verifiable step.”
Other / Unspecified	$N = 215$	7.4%	<i>unspecified, error, incorrect</i>	“Vague failure critique without explicit grounding attribution.”
Perceptual Total	$N = 1,270$	43.5%	<i>Combined Axis Misread (24.0%) and Series/Legend Identity (19.5%)</i>	
Arithmetic Total	$N = 38$	1.3%	<i>Pure calculation failure accounts for only 1 in 77 step errors</i>	

Table 5: **PRM Judge Error Taxonomy from 2,920 Refuted Reasoning Steps.** Categorized via priority-regex analysis over natural language critiques from muse-spark-1.1. Visual grounding (axes, series colors) forms the primary reasoning bottleneck, while arithmetic mistakes are rare.

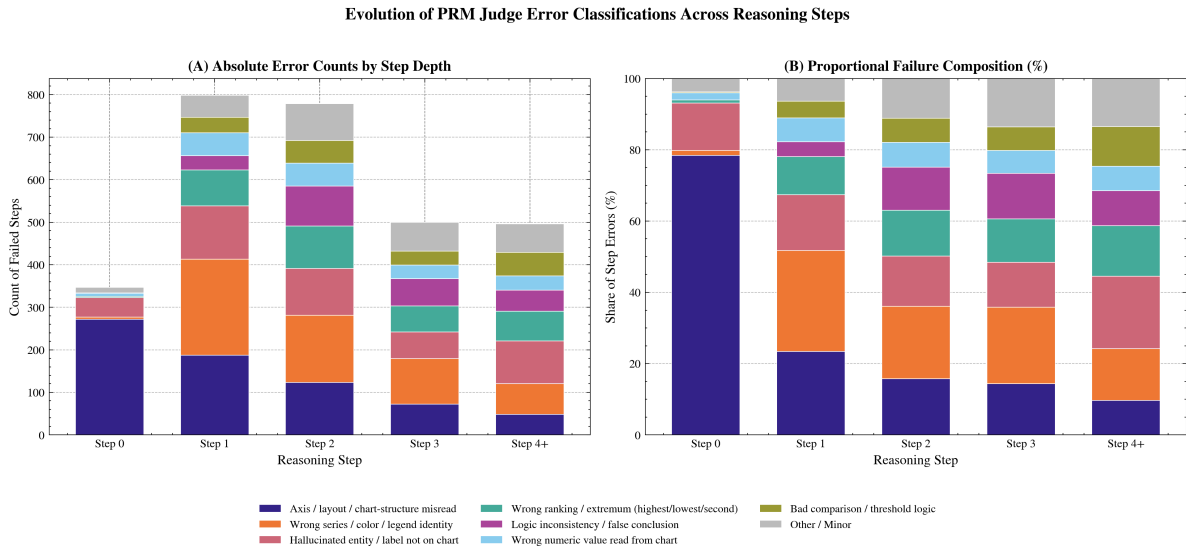


Figure 5: **Error Mode Composition Across Step Depth.** Detailed breakdown of 2,920 refuted reasoning steps evaluated by muse-spark-1.1 across reasoning steps 0 to 4+. Step 0 is dominated by axis and layout misreads (78.5%), which triggers cascading entity hallucinations and logical inconsistencies in downstream steps.

Trajectory Metric	Count	Share
CharXiv Reasoning Problems	500	100.0%
Evaluated Rollout Traces	1,274	100.0%
Mean Candidates / Question	3.83	–
Total Judged Reasoning Steps	4,947	100.0%
Step Verification:		
– Step Pass Rate (Score = 1)	2,027	41.0%
– Step Failure Rate (Score = 0)	2,920	59.0%
Rollout Outcome:		
– Fully Valid Rollouts (All steps = 1)	154	12.1%
– Rollouts with ≥ 1 Step Error	1,120	87.9%
Mean Steps per Rollout	3.88	–
Failure Progression Dynamics:		
– First Error at Step 0	347	31.0%
– First Error at Step 1	546	48.8%
– First Error at Step ≤ 1 Cumulative	893	79.7%
– Cascade: $P(S_{N+1} = 0 \mid S_N = 0)$	1,630	82.7%
– Recovery: $P(S_{N+1} = 1 \mid S_N = 0)$	340	17.3%

Table 6: **Reasoning Trajectory Statistics on CharXiv 500 Pool.** Over 79% of trajectory errors begin within the first two steps. Once an error occurs, downstream steps fail 82.7% of the time.

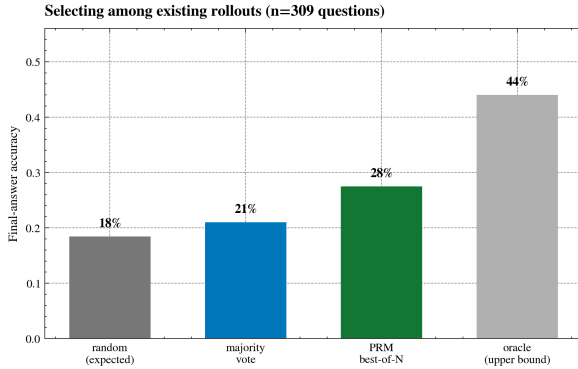


Figure 6: **Inference-Time PRM Verifier Accuracy.** Average step process reward selection (27.5%) outperforms self-consistency majority voting (21.0%) and random rollouts (18.4%) across 309 multi-candidate questions.

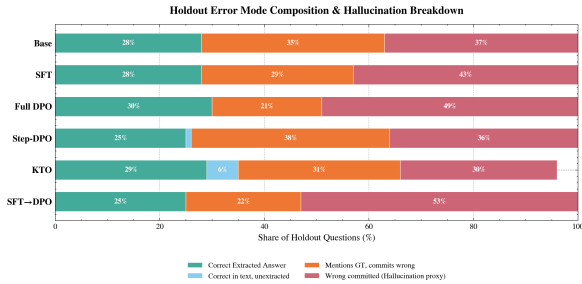


Figure 7: **Holdout Error Mode Breakdown.** Partitioning of test responses into extracted correct, unextracted correct in prose, partial ground-truth mentions, and committed wrong answers (hallucination proxy).

Training Dynamics & Loss Trajectories Across Fine-Tuning Paradigms

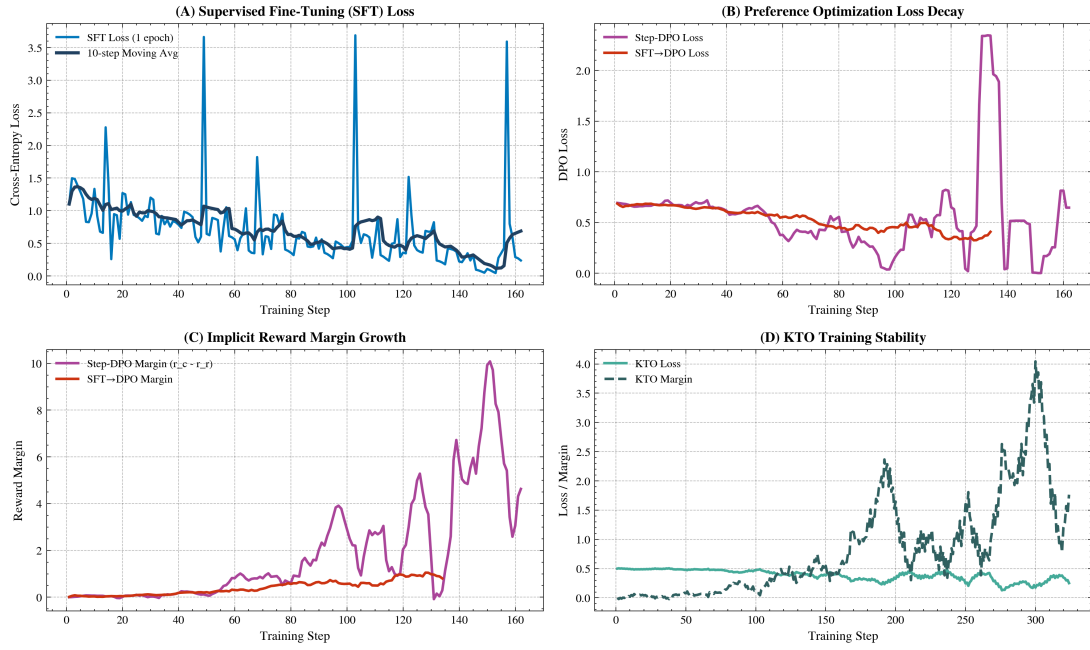


Figure 8: **Training Dynamics Across Alignment Paradigms.** Top-left: SFT cross-entropy loss decay. Top-right: DPO loss trajectories. Bottom-left: Step-DPO implicit reward margin explosion ($\Delta r \rightarrow +10.0$). Bottom-right: KTO margin oscillations under single-T4 GPU constraints.

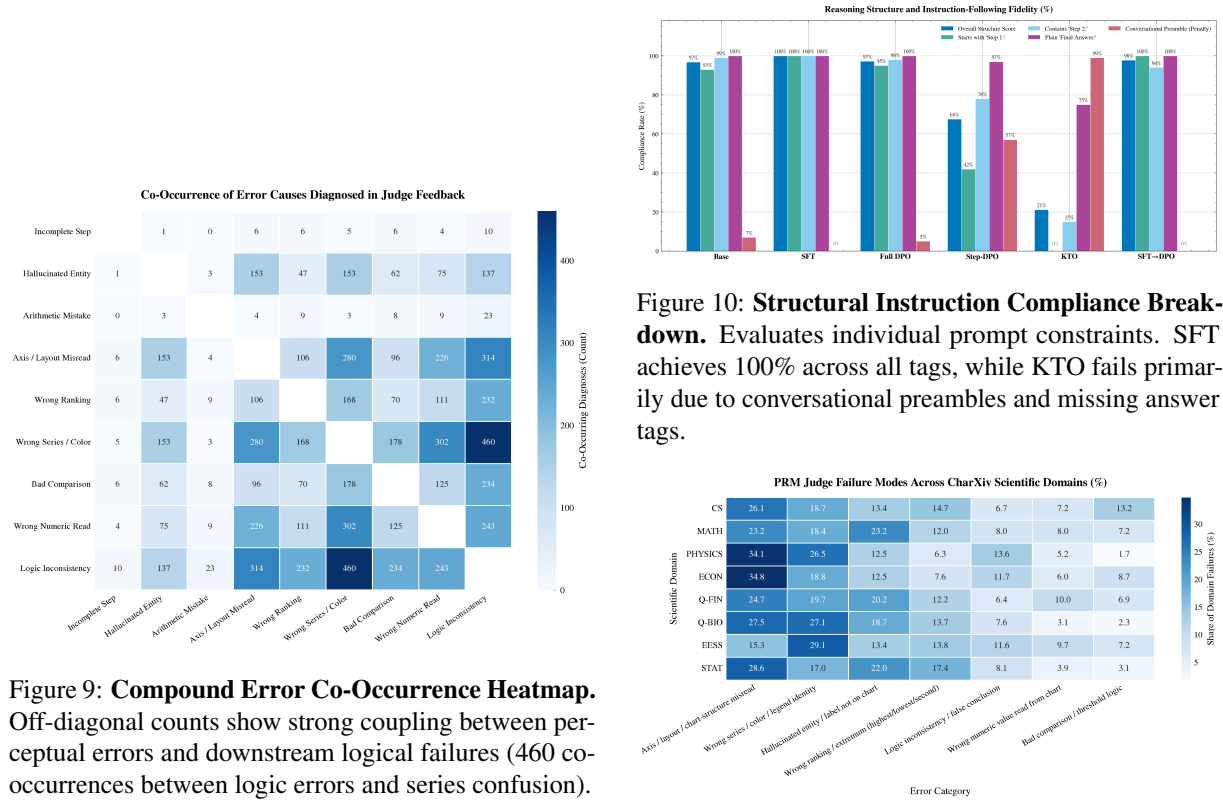


Figure 9: **Compound Error Co-Occurrence Heatmap.** Off-diagonal counts show strong coupling between perceptual errors and downstream logical failures (460 co-occurrences between logic errors and series confusion).

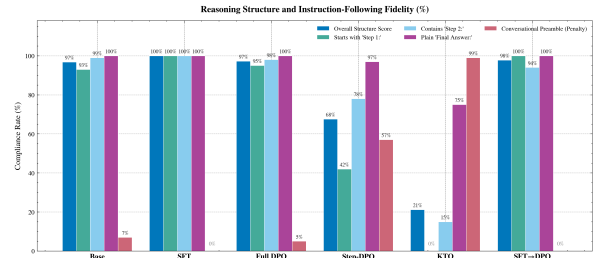


Figure 10: **Structural Instruction Compliance Breakdown.** Evaluates individual prompt constraints. SFT achieves 100% across all tags, while KTO fails primarily due to conversational preambles and missing answer tags.

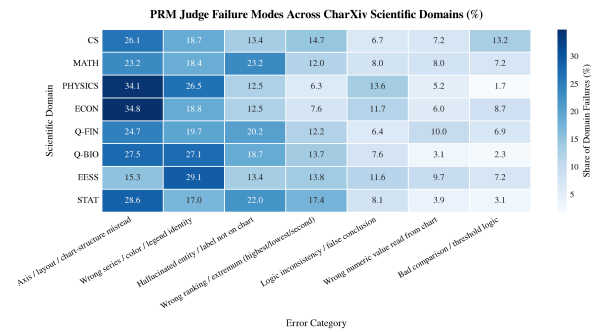


Figure 11: **Error Distribution Across CharXiv Domains.** Physics and Economics charts exhibit the highest rate of axis misreads (>34%), while Mathematics and Computer Science suffer more from entity hallucinations (>22%).

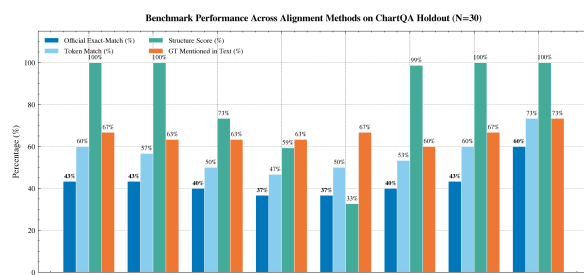


Figure 12: **ChartQA Holdout Benchmark** ($N = 30$). Pareto-DPO leads on exact-match, token match, and structure score simultaneously, matching the pattern reported numerically in Table 4.